

# Observational Evidence of Kelvin Waves in the Tropical Stratosphere<sup>1</sup>

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## ABSTRACT

This study of synoptic-scale wave motions in the equatorial stratosphere is based on the analysis of six months of radiosonde data from three tropical stations. Fluctuations in the zonal wind with an average period of 15 days and amplitudes in the order of  $8\text{--}12\text{ m sec}^{-1}$  are noted. Corresponding fluctuations are found in the temperature field with amplitudes of  $3\text{--}5^\circ\text{C}$  and a phase lead of  $\frac{1}{2}$  cycle with respect to the zonal wind. These wave motions which propagate phase downward do not appear to involve the meridional wind component.

The observed fluctuations resemble Kelvin waves, which represent one of the solutions of the wave equation on an equatorial beta plane. One of the notable features of this type of wave is that it produces an upward flux of westerly momentum. The observations indicate that this flux is large enough to account for the westerly accelerations associated with the quasi-biennial oscillation.

## 1. Introduction

This study of synoptic-scale disturbances in the tropical stratosphere was originally motivated by certain unsolved problems relating to the momentum budget of the quasi-biennial oscillation. Wallace and Holton (1968) noted that the downward propagation of alternating easterly and westerly wind regimes without loss of amplitude, which is observed in the  $20\text{--}30\text{ km}$  layer, could be explained only on the basis of a downward propagating source of zonal momentum. It was suggested by Lindzen (personal communication) that the interaction between vertically propagating wave disturbances and the zonal flow might provide just such a momentum source. The upward flux of momentum in wave disturbances might be trapped in the layers of strong vertical wind shear which separate the easterly and westerly wind regimes, and absorption of this momentum could, in turn, cause the shear layers to move downward. This idea was based on Booker and Bretherton's (1967) theoretical study of the trapping of vertically propagating disturbances in layers of strong vertical wind shear.

For such a mechanism to be effective, four conditions would have to be met:

- 1) Wave disturbances would have to be present in the tropical stratosphere, at least at the times when the wind regimes exhibit downward propagation.
- 2) The structure of these disturbances would have to result in an upward flux of zonal momentum of the same sign as the advancing wind regime. For example, a westerly regime could propagate downward to replace an easterly regime only by means

of absorption of westerly momentum in the intervening layer.

- 3) The disturbances would have to be of sufficient intensity to account for the observed accelerations.
- 4) The east-west wave velocities of the disturbances would have to have roughly the same range as the zonal wind speeds encountered within the wind

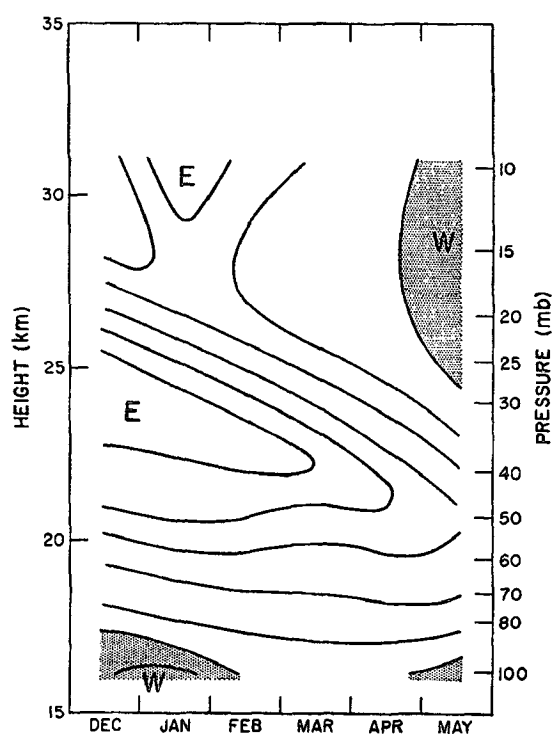


FIG. 1. Time-height section of zonal wind at  $8^\circ$  latitude. Data from Balboa ( $9^\circ\text{N}$ ) and Ascension Island ( $7^\circ\text{S}$ ) have been averaged together in order to eliminate the annual cycle. Isotachs are placed at intervals of  $5\text{ m sec}^{-1}$ . Westerlies are shaded.

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regimes. This is a necessary condition for absorption of the waves by the zonal flow.

## 2. Subjective analysis of the data

In order to test Lindzen's hypothesis a synoptic study was undertaken using twice daily radiosonde data for levels above 100 mb from three stations—Balboa (9N, 80W), Trinidad (11N, 61W), and Kwajalein (9N, 168E) for the six month period 1 December 1965 through 31 May 1966. Fig. 1 shows that during this

time a westerly regime was propagating downward through the 15–30 mb layer, replacing rather strong easterlies.

Figs. 2–6 show subjectively analyzed time-height sections of zonal wind and temperature during this period. The sections show fluctuations with periods on the order of 10–15 days and average amplitudes of 8–12 m sec<sup>-1</sup> and 3–5°C, respectively. The strong similarity between the wind fluctuations at Balboa and Trinidad, 20° of longitude apart, is evidence of the large scale of these

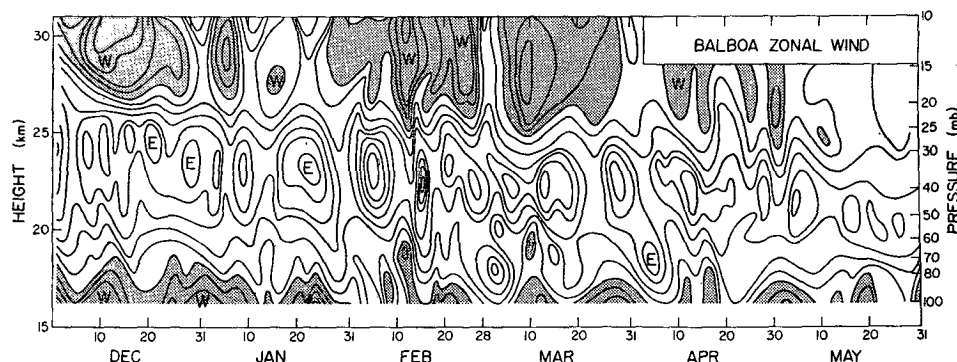


FIG. 2. Time-height section of zonal wind at Balboa. Isotachs are placed at intervals of 5 m sec<sup>-1</sup>. Westerlies are shaded.

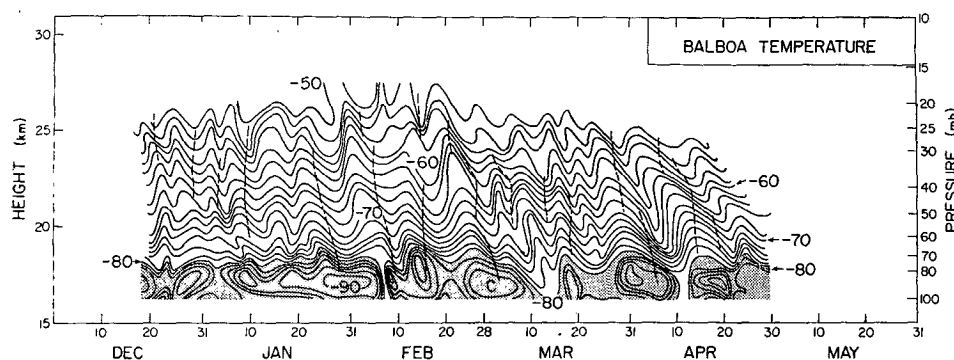


FIG. 3. Time-height section of temperature at Balboa. Isotherms are placed at intervals of 2°C. Temperatures below -80°C are shaded. Dashed lines represent axes of the more prominent easterly fluctuations shown in Fig. 2.

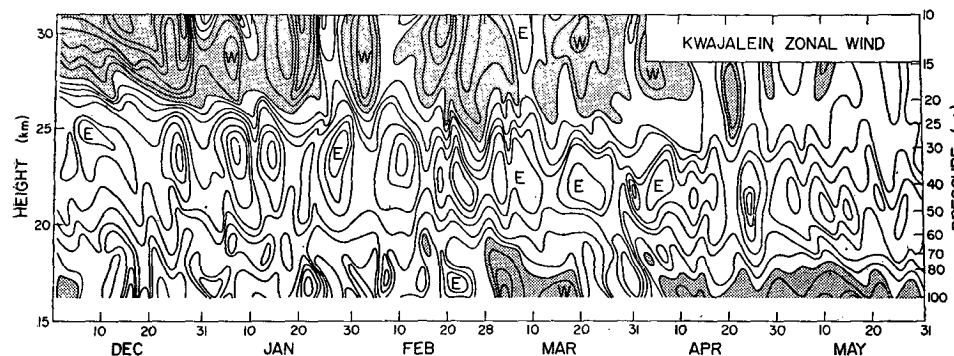


FIG. 4. Time-height section of zonal wind at Kwajalein. Isotachs are placed at intervals of 5 m sec<sup>-1</sup>. Westerlies are shaded.

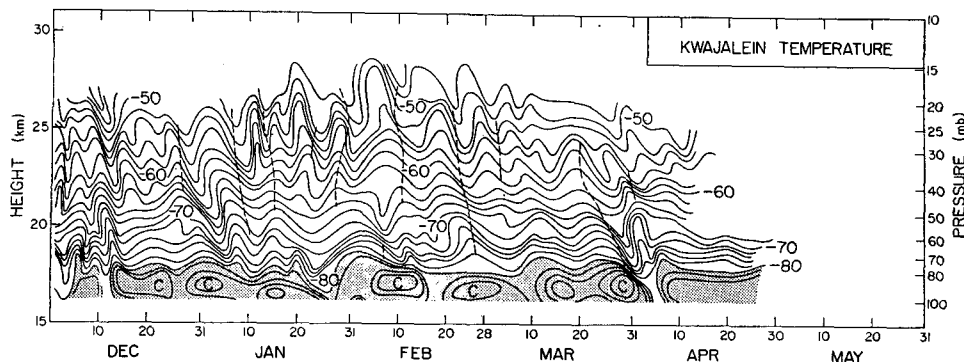


FIG. 5. Time-height section of temperature at Kwajalein. Isotherms are placed at intervals of  $2^{\circ}\text{C}$ . Temperatures below  $-80^{\circ}\text{C}$  are shaded. Dashed lines represent axes of the more prominent easterly fluctuations shown in Fig. 4.

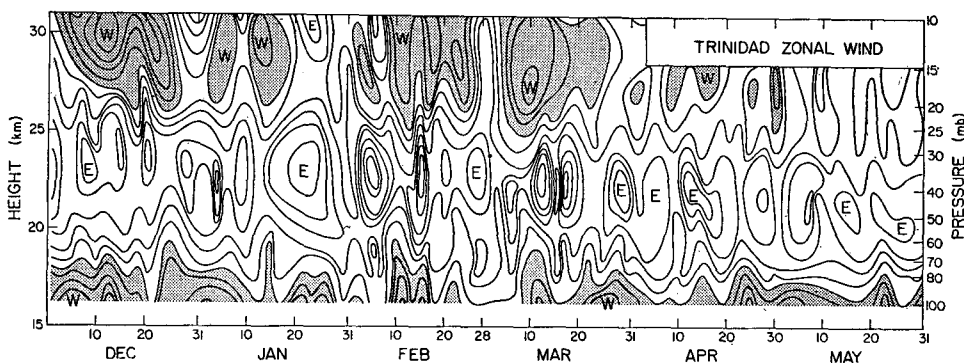


FIG. 6. Time-height section of zonal wind at Trinidad. Isotachs are placed at intervals of  $5 \text{ m sec}^{-1}$ . Westerlies are shaded.

features. However, the large phase differences between Balboa and Kwajalein establish the fact that the motions are wavelike rather than axially symmetric in nature. A phase relationship is apparent between the wind and temperature fluctuations at individual stations. Easterly wind maxima, indicated by dashed lines on the temperature sections, tend to be situated in regions of high static stability, about  $\frac{1}{4}$  wavelength in advance of the temperature maxima. Both wind and temperature oscillations show evidence of downward propagation, the latter in particular.

Meridional wind data are not shown in the sections because the fluctuations are quite small (on the order of  $2\text{--}3 \text{ m sec}^{-1}$ ) and show little evidence of organization of any type.

### 3. Similarities between the observed features and Kelvin waves

Matsuno (1966) showed that horizontally propagating Kelvin waves are a possible free solution to the perturbation equations written for an equatorial beta plane in a homogeneous, incompressible fluid. Although no free Kelvin waves exist in an unbounded atmosphere, Holton and Lindzen (1968) have demonstrated the existence of a forced vertically propagating mode similar in structure to the barotropic Kelvin wave. The solutions

yield a latitudinal distribution of amplitude which has the shape of a probability curve centered at the equator. In order to satisfy boundary conditions the waves must

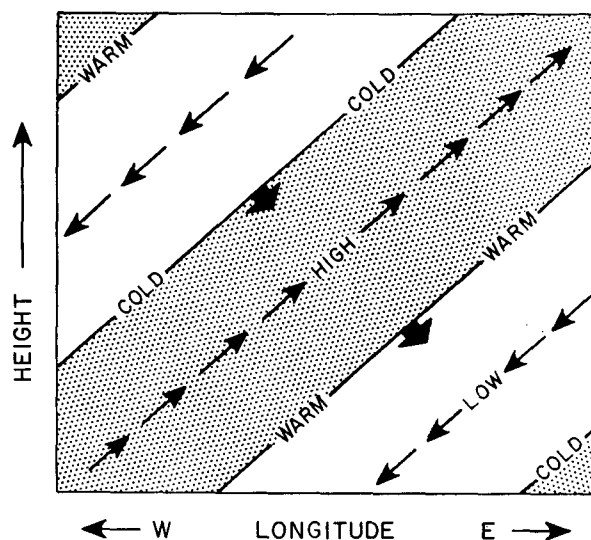


FIG. 7. Idealized cross section along a latitude circle showing phases of the zonal wind, temperature, pressure, and vertical motion oscillations associated with Kelvin waves, as deduced from theoretical considerations.

propagate eastward (Holton and Lindzen) and, in the atmospheric case, downward (Lindzen, 1967).

The wind, pressure and temperature relationships within Kelvin waves, sketched in Fig. 7, can be completely derived from theoretical considerations:

1) The axes of maximum upward motion and maximum westerly wind must coincide. This is the only arrangement which satisfies the continuity equation in the absence of a meridional wind component. It results in an upward flux of westerly momentum.

2) The zonal wind and temperature oscillations must be in quadrature in order to satisfy the thermal wind equation for the zonal wind component. The temperature maximum precedes the westerly wind maximum by  $\frac{1}{4}$  wavelength.

3) The hydrostatic equation requires that the pressure oscillation lag  $\frac{1}{4}$  wavelength behind the temperature

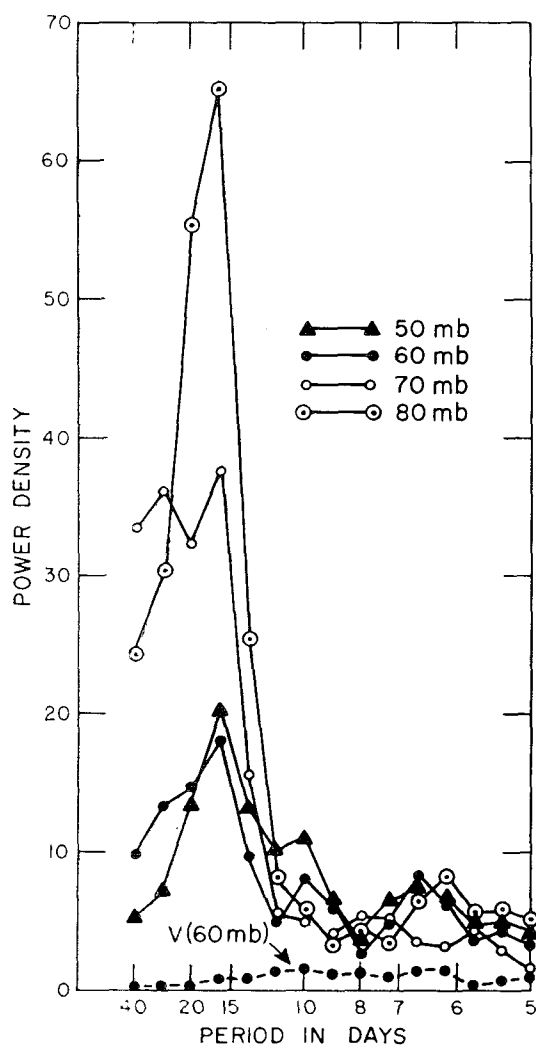


FIG. 8. Power spectra of zonal wind fluctuations at 80, 70, 60 and 50 mb shown together with the spectrum for the meridional wind fluctuations at 60 mb. Zonal wind data are prewhitened by means of a procedure described in the text.

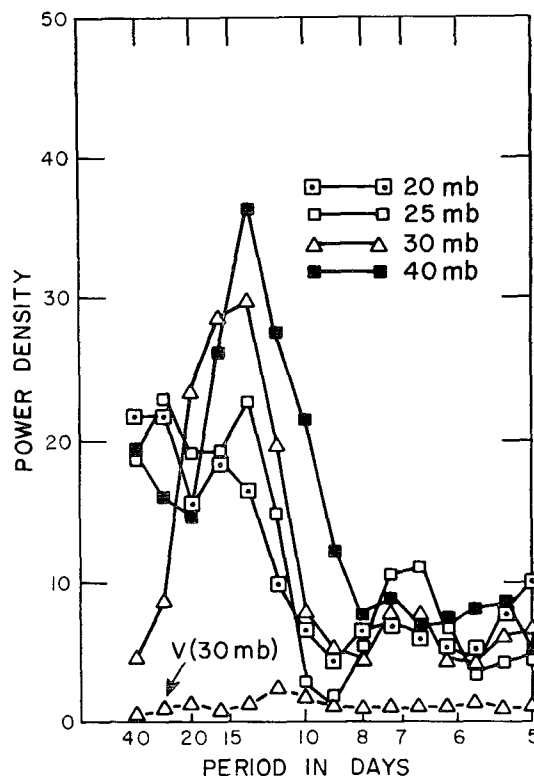


FIG. 9. Power spectra of zonal wind fluctuations at 40, 30, 25 and 20 mb shown together with the spectrum for the meridional wind fluctuations at 30 mb. Zonal wind data are prewhitened by means of a procedure described in the text.

oscillation. Thus, the zonal wind and pressure waves coincide, resulting in an upward flux of wave energy.

In this arrangement, the peak downward motion occurs  $\frac{1}{4}$  cycle in advance of warmest temperature, i.e., at the time when the temperature is increasing most rapidly. The peak east-to-west pressure gradient is felt  $\frac{1}{4}$  cycle in advance of the westerly wind maximum, i.e., when the largest westerly accelerations are taking place. Thus, the model is physically consistent so long as diabatic heating is small relative to the adiabatic effects.

The wind and temperature analyses shown in Figs. 2-6 appear to be consistent with this model; downward propagation is observed, and the wind and temperature fluctuations show the proper phase relationship. This, together with the absence of organized meridional wind fluctuations, strongly suggests that the more pronounced features in the data are associated with the passage of Kelvin waves.

#### 4. Objective analysis of the data

In order to objectively verify the phase relationships suggested by Figs. 2-6 and to test for some of the other properties of Kelvin waves, the data were subjected to power spectrum analysis. Time series were prepared from the twice daily wind, height and temperature data at each reporting level. The zonal wind data were pre-

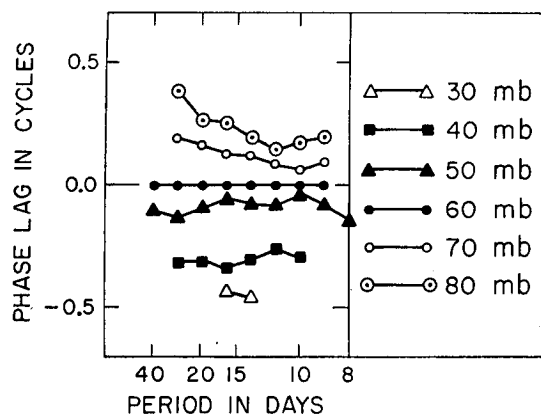


FIG. 10. Phase spectra of temperature at various levels with respect to 60 mb. Positive values indicate phase lags. The corresponding coherence squares are shown in Table 1.

whitened by generating a second series, composed of linear interpolations of the monthly mean values, and subtracting this from the original series. This procedure should have little effect on fluctuations with time scales of a month or less. The full six-month period was analyzed using a lag period of 40 days.

Figs. 8 and 9 show the power spectra of zonal wind for 80, 70, 60, 50, 40, 30, 25 and 20 mb at Balboa. Peaks near the period of 15 days are prominent in all the curves with largest values occurring at 80 and 30 mb. By contrast, the spectra for the meridional component at 60 and 30 mb, shown in the same diagrams, do not exhibit any notable peaks in this frequency range.

Fig. 10 shows the phase spectra of the Balboa data with respect to the 60-mb temperature series. Downward propagation is evident over a wide range of frequencies, with a wavelength of about 10 km. Phase spectra of the zonal wind data, shown in Fig. 11, are also indicative of downward phase propagation for periods near 15 days. However, in contrast to the temperature data, wind fluctuations with periods of less than 10

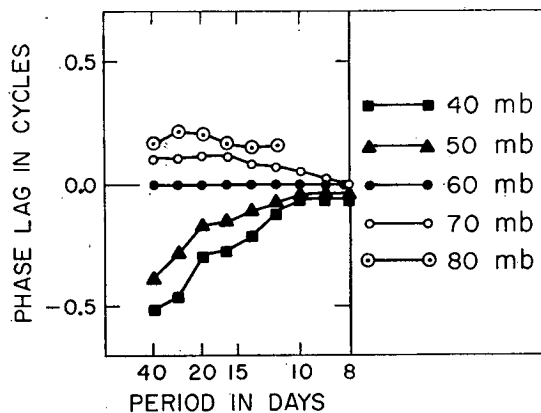


FIG. 11. Phase spectra of zonal wind at various levels with respect to 60 mb. Positive values indicate phase lags. The corresponding coherence squares are shown in Table 2.

TABLE 1. Coherence square between temperatures at 60 mb and adjacent levels as a function of period.

| Level (mb) | Period (days) |      |      |      |      |      |      |      |      |
|------------|---------------|------|------|------|------|------|------|------|------|
|            | 40.0          | 26.7 | 20.0 | 16.2 | 13.3 | 11.4 | 10.0 | 8.9  | 8.0  |
| 30         | 0.11          | 0.04 | 0.09 | 0.36 | 0.34 | 0.04 | 0.21 | 0.53 | 0.28 |
| 40         | 0.17          | 0.27 | 0.48 | 0.79 | 0.63 | 0.46 | 0.28 | 0.02 | 0.10 |
| 50         | 0.56          | 0.59 | 0.68 | 0.90 | 0.79 | 0.72 | 0.58 | 0.33 | 0.46 |
| 60         | 1.00          | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 |
| 70         | 0.19          | 0.60 | 0.76 | 0.90 | 0.91 | 0.74 | 0.44 | 0.12 | 0.26 |
| 80         | 0.12          | 0.30 | 0.46 | 0.69 | 0.59 | 0.57 | 0.59 | 0.40 | 0.07 |

TABLE 2. Coherence square between zonal wind at 60 mb and adjacent levels as a function of period.

| Level (mb) | Period (days) |      |      |      |      |      |      |      |      |
|------------|---------------|------|------|------|------|------|------|------|------|
|            | 40.0          | 26.7 | 20.0 | 16.2 | 13.3 | 11.4 | 10.0 | 8.9  | 8.0  |
| 40         | 0.66          | 0.51 | 0.59 | 0.45 | 0.21 | 0.33 | 0.45 | 0.45 | 0.26 |
| 50         | 0.52          | 0.46 | 0.75 | 0.82 | 0.60 | 0.47 | 0.60 | 0.64 | 0.55 |
| 60         | 1.00          | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 |
| 70         | 0.85          | 0.83 | 0.80 | 0.87 | 0.78 | 0.84 | 0.77 | 0.54 | 0.32 |
| 80         | 0.62          | 0.62 | 0.63 | 0.72 | 0.50 | 0.35 | 0.21 | 0.21 | 0.04 |

days are practically simultaneous at all levels. This accounts for the somewhat different appearance of the time-height sections.

Tables of the coherence squares corresponding to the data points plotted in Figs. 10 and 11 are shown for reference in Tables 1 and 2. The coherences are generally high for adjacent levels. Phase data for which the coherences are consistently below 0.3 have been omitted.

Fig. 12 shows the wind-temperature phase spectra at Balboa for the full range of levels. With few exceptions the data points are situated very close to the theoretical value of  $\frac{1}{4}$  wavelength. The fact that 80 mb is the average location of the tropopause where the static stability undergoes a large change may explain the poor fit at that level. The corresponding coherence squares are shown in Table 3. Data points with coherence squares of less than 0.3 are not plotted in the figure.

Fig. 13 is an attempt to relate the phases of the wind and height fluctuations at Balboa to one another. The

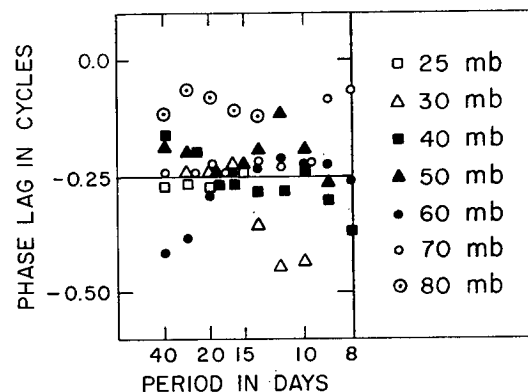


FIG. 12. Phase spectra of temperature with respect to zonal wind at all levels between 80 and 25 mb. Positive values indicate phase lags. The corresponding coherence squares are shown in Table 3. The theoretical value of  $-\frac{1}{4}$  cycle is shown for comparison.

TABLE 3. Coherence square between zonal wind and temperature as a function of period.

| Level<br>(mb) | Period (days) |      |      |      |      |      |      |      |      |
|---------------|---------------|------|------|------|------|------|------|------|------|
|               | 40.0          | 26.7 | 20.0 | 16.2 | 13.3 | 11.4 | 10.0 | 8.9  | 8.0  |
| 20            | 0.05          | 0.34 | 0.43 | 0.24 | 0.10 | 0.19 | 0.27 | 0.11 | 0.15 |
| 25            | 0.02          | 0.41 | 0.68 | 0.34 | 0.02 | 0.04 | 0.00 | 0.07 | 0.02 |
| 30            | 0.00          | 0.59 | 0.85 | 0.66 | 0.35 | 0.61 | 0.51 | 0.17 | 0.50 |
| 40            | 0.47          | 0.71 | 0.62 | 0.39 | 0.17 | 0.30 | 0.47 | 0.28 | 0.22 |
| 50            | 0.61          | 0.37 | 0.56 | 0.76 | 0.23 | 0.27 | 0.45 | 0.49 | 0.08 |
| 60            | 0.32          | 0.31 | 0.43 | 0.89 | 0.75 | 0.65 | 0.57 | 0.45 | 0.07 |
| 70            | 0.57          | 0.84 | 0.63 | 0.58 | 0.37 | 0.35 | 0.42 | 0.30 | 0.51 |
| 80            | 0.35          | 0.49 | 0.52 | 0.60 | 0.29 | 0.04 | 0.19 | 0.31 | 0.42 |

TABLE 4. Coherence square between zonal wind and height as a function of period.

| Level<br>(mb) | Period (days) |      |      |      |      |      |      |      |      |
|---------------|---------------|------|------|------|------|------|------|------|------|
|               | 40.0          | 26.7 | 20.0 | 16.2 | 13.3 | 11.4 | 10.0 | 8.9  | 8.0  |
| 30            | 0.04          | 0.08 | 0.17 | 0.23 | 0.07 | 0.02 | 0.10 | 0.00 | 0.16 |
| 40            | 0.26          | 0.78 | 0.07 | 0.03 | 0.10 | 0.20 | 0.09 | 0.01 | 0.05 |
| 50            | 0.14          | 0.78 | 0.08 | 0.23 | 0.35 | 0.19 | 0.05 | 0.04 | 0.06 |
| 60            | 0.01          | 0.56 | 0.03 | 0.47 | 0.67 | 0.30 | 0.03 | 0.16 | 0.21 |
| 70            | 0.04          | 0.00 | 0.17 | 0.52 | 0.61 | 0.10 | 0.01 | 0.36 | 0.17 |
| 80            | 0.02          | 0.25 | 0.17 | 0.36 | 0.43 | 0.11 | 0.02 | 0.36 | 0.08 |

corresponding coherence squares shown in Table 4 are not impressively large, probably because of the high noise level inherent in the height data. Nevertheless, there appears to be some clustering of the data points near the zero line, as predicted by the theory. This is particularly noticeable near 15-day periods and at the lower levels, where the coherences appear to be significant. Several years of data would be required to firmly establish this relationship.

Fig. 14 shows the phase relationship between the zonal wind oscillations at Balboa (80W) and Trinidad (61W). In the period range of 10–20 days, the fluctuations at Trinidad lag slightly behind those at Balboa at 80, 70, 60 and 50 mb and lead slightly at the higher levels. This may be regarded as evidence of eastward propagation below 40 mb and westward propagation above.

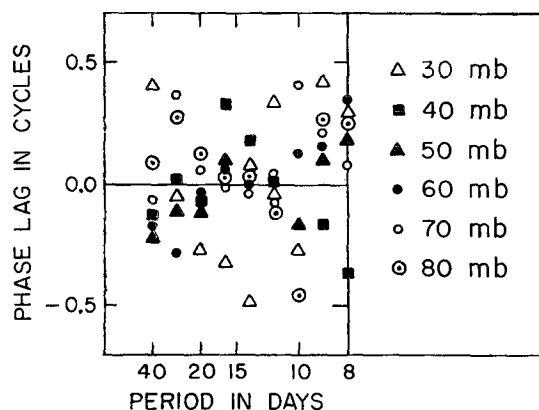


FIG. 13. Phase spectra of geopotential height with respect to zonal wind for each level between 80 and 30 mb. Positive values indicate phase lags. The corresponding coherence squares are shown in Table 4.

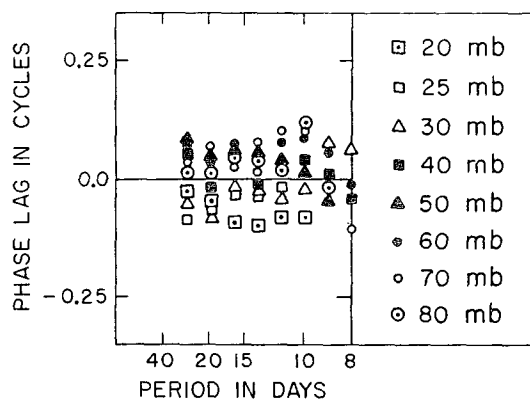


FIG. 14. Phase spectra of zonal wind at Trinidad with respect to Balboa for each level between 80 and 20 mb. Positive values indicate phase lags. The corresponding coherence squares are shown in Table 5.

The small phase differences, generally less than 0.1 cycle, are suggestive of a zonal wavelength in the range of wavenumber one. This agrees with the theoretical calculations of Holton and Lindzen (1968). The corresponding coherence squares shown in Table 5 are quite high, as would be expected from the time-height sections. The phase spectra for Balboa and Kwajalein (162E) shown in Fig. 15 suggest lags near  $\frac{1}{2}$  cycle, which is indicative of somewhat shorter wavelengths and a change in the shape of the waves with longitude. It is clear that data from several additional stations will be needed to resolve the longitudinal structure and movement of the waves.

The consistency between the observed oscillation and the model shown in Fig. 7 is quite good, with only two exceptions; at 80 mb the phase lag between zonal wind

TABLE 5. Coherence square between zonal winds at Balboa and Trinidad as a function of period.

| Level<br>(mb) | Period (days) |      |      |      |      |      |      |      |      |
|---------------|---------------|------|------|------|------|------|------|------|------|
|               | 40.0          | 26.7 | 20.0 | 16.2 | 13.3 | 11.4 | 10.0 | 8.9  | 8.0  |
| 20            | 0.77          | 0.74 | 0.58 | 0.71 | 0.77 | 0.62 | 0.45 | 0.09 | 0.12 |
| 25            | 0.57          | 0.86 | 0.82 | 0.76 | 0.79 | 0.79 | 0.11 | 0.09 | 0.20 |
| 30            | 0.28          | 0.63 | 0.86 | 0.89 | 0.94 | 0.80 | 0.68 | 0.69 | 0.49 |
| 40            | 0.48          | 0.51 | 0.85 | 0.83 | 0.75 | 0.71 | 0.78 | 0.55 | 0.52 |
| 50            | 0.74          | 0.76 | 0.88 | 0.86 | 0.67 | 0.64 | 0.74 | 0.69 | 0.55 |
| 60            | 0.87          | 0.78 | 0.60 | 0.66 | 0.68 | 0.35 | 0.53 | 0.47 | 0.26 |
| 70            | 0.64          | 0.75 | 0.63 | 0.63 | 0.52 | 0.51 | 0.59 | 0.28 | 0.11 |
| 80            | 0.35          | 0.49 | 0.45 | 0.48 | 0.43 | 0.51 | 0.29 | 0.01 | 0.06 |

TABLE 6. Coherence square between zonal winds at Balboa and Kwajalein as a function of period.

| Level<br>(mb) | Period (days) |      |      |      |      |      |      |      |      |
|---------------|---------------|------|------|------|------|------|------|------|------|
|               | 40.0          | 26.7 | 20.0 | 16.2 | 13.3 | 11.4 | 10.0 | 8.9  | 8.0  |
| 20            | 0.27          | 0.61 | 0.01 | 0.15 | 0.00 | 0.10 | 0.04 | 0.07 | 0.54 |
| 25            | 0.18          | 0.49 | 0.54 | 0.36 | 0.36 | 0.29 | 0.06 | 0.22 | 0.23 |
| 30            | 0.03          | 0.69 | 0.57 | 0.24 | 0.30 | 0.41 | 0.25 | 0.24 | 0.08 |
| 40            | 0.30          | 0.08 | 0.22 | 0.22 | 0.25 | 0.34 | 0.21 | 0.19 | 0.21 |
| 50            | 0.09          | 0.00 | 0.17 | 0.41 | 0.39 | 0.06 | 0.10 | 0.09 | 0.08 |
| 60            | 0.35          | 0.37 | 0.36 | 0.38 | 0.36 | 0.09 | 0.11 | 0.28 | 0.26 |
| 70            | 0.29          | 0.42 | 0.44 | 0.59 | 0.48 | 0.12 | 0.40 | 0.40 | 0.20 |
| 80            | 0.35          | 0.23 | 0.53 | 0.62 | 0.34 | 0.05 | 0.24 | 0.41 | 0.67 |

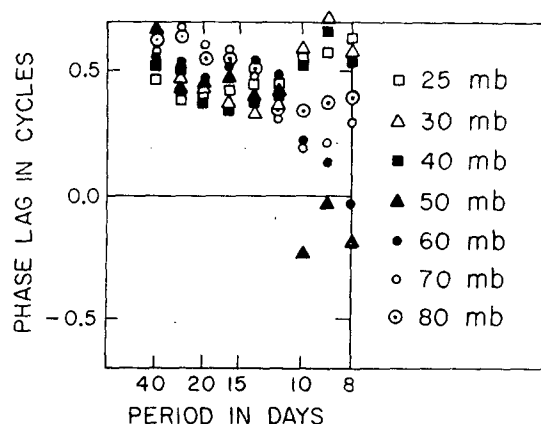


FIG. 15. Phase spectra of zonal wind at Kwajalein with respect to Balboa for each level between 80 and 20 mb. Positive values indicate phase lags. The corresponding coherence squares are shown in Table 6.

and temperature is somewhat less than indicated by theory but, as noted above, the proximity of this level to the tropopause may tend to distort the temperature field within the waves. At levels above 40 mb Fig. 14 suggests westward phase propagation, which is contrary to the model. This behavior may be peculiar to this limited data sample. However, it should be noted that the subject of Kelvin waves in a shear flow has not yet been treated theoretically and, therefore, we should not preclude the possibility of westward phase propagation in regions of strong vertical shear. Fig. 1 shows that quite strong shear is prevalent above 40 mb throughout the period.

## 5. Discussion

From the above discussion it can be concluded that there are wave disturbances in the tropical stratosphere which exhibit many of the properties of atmospheric Kelvin waves. We now consider the question of whether such waves might provide the vehicle for vertical momentum transport in the mechanism suggested by Lindzen for explaining the zonal wind changes associated with the quasi-biennial oscillation. Let us re-examine the criteria set forth in the introduction:

- 1) If the mechanism suggested by Lindzen was responsible for the changes in the zonal wind profile during the period in question, large upward fluxes of westerly momentum must have been occurring. It has been shown that fluctuations which resemble Kelvin waves were present and it appears likely that they were producing an upward flux of westerly momentum. Whether these waves always accompany the descent of westerly regimes is not yet known.
- 2) It is clear from the foregoing discussion that Kelvin waves cannot transport easterly momentum upward. Therefore, these waves cannot be called

upon to explain the descent of easterly regimes, unless it turns out that for some reason they are confined to regions of mean easterlies. If this were the case, there would be a divergence of westerly momentum near the bases of easterly regimes and this would cause them to propagate downward. In a sense, this would be the reverse of Lindzen's mechanism, with the waves absorbing westerly momentum from the mean flow.

- 3) During the period in question, according to Fig. 1, the mean zonal wind in the 15–30 mb layer changed from about  $25 \text{ m sec}^{-1}$  easterly to about zero. (Fig. 1 is based on averages of Balboa and Ascension Island data so as to exclude the effect of the annual cycle, which is presumably caused by mean meridional motions.) This wind change corresponds to a mean westerly acceleration of about  $2 \times 10^{-6} \text{ m sec}^{-2}$ .

The observed waves exhibit a mean period of about 15 days and amplitudes of about  $8 \text{ m sec}^{-1}$  for the zonal wind and  $3\text{C}$  for the temperature. These conservative estimates imply a vertical motion oscillation of about  $0.1 \text{ cm sec}^{-1}$ . The resemblance between the observed fluctuations and Kelvin waves would lead one to expect a high correlation between the zonal and vertical motions. Assuming a correlation coefficient of 0.5 and complete absorption of the momentum by the zonal wind in the 15–30 mb layer leads to an estimated acceleration which is about the same size as that quoted above. Thus, the waves described above should be of sufficient intensity to account for the accelerations observed during the descent of westerly regimes.

- 4) Since the data thus far examined do not clearly define the longitudinal scale of the disturbances, the question of phase speeds cannot yet be resolved. If, as is indicated by the data, the waves change shape as they move longitudinally, this may prove to be a difficult problem.

Although present indications are encouraging, it is clear that further evidence will be needed to firmly establish a cause-and-effect relationship between Kelvin waves and the biennial oscillation. It would be particularly desirable to obtain information on the time variations in amplitude and structure of the waves over a period of one or more cycles of the biennial oscillation. A more detailed study of the structure and phase propagation of the waves utilizing data from many more stations would also be useful.

It should be noted that the wave motions identified in this study are distinctly different from those found by Yanai and Maruyama (1966), and discussed by Maruyama and Yanai (1967) and Maruyama (1967). Table 7 summarizes the characteristics of the two types. The latter type of disturbance resembles another of the

TABLE 7. Comparison of wave motion features identified in this study with those found by Yanai and Maruyama (1966).

|                   | Wallace and Kousky       | Yanai and Maruyama      |
|-------------------|--------------------------|-------------------------|
| Mean period       | 15 days                  | 5 days                  |
| Phase propagation | downward, eastward       | downward, westward      |
| Type of motion    | zonal and vertical only  | three dimensional       |
| Amplitudes        |                          |                         |
| Zonal motion      | 8–12 m sec <sup>-1</sup> | 2 m sec <sup>-1</sup>   |
| Meridional motion | 0                        | 2–3 m sec <sup>-1</sup> |
| Temperature       | 3–5°C                    | 2–3°C                   |

solutions of the wave equation for an equatorial beta plane (Lindzen and Matsuno, 1968).

The possibility of a third type of motion in this region arises from a study of 10-mb wind data by Riehl and Higgs (1960). During a five day period they followed a disturbance resembling a shear line as it propagated northward across the Caribbean region. However, it is also possible that the phenomenon which they described may have been related to the passage of either of the types of waves described above. Although the theory does not lead us to expect meridional phase propagation of these waves, it is possible that interactions with the meridional shear of the mean zonal flow might produce such effects.

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